CSCI 544: Applied Natural Language Processing

Transformer-II

Xuezhe Ma (Max)

USC University of Southern California

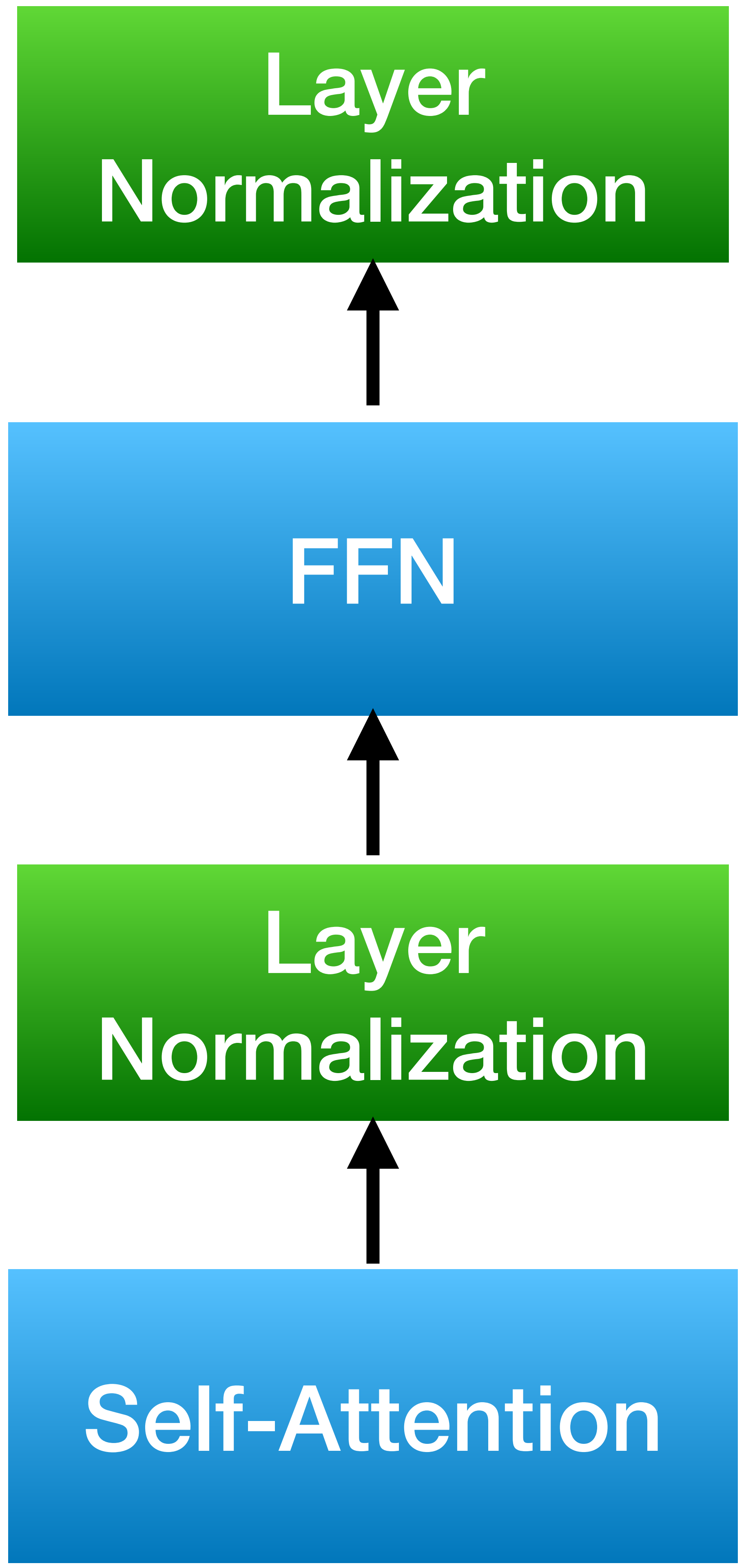
Course Organization

NLP Tasks

Models	Text Classification	Sequence Labeling	Seq2seq & MT	Language Modeling & Generation	Advanced Topics
	Naive Bayes Logistic Regression SVM …	MLP CNN	……	……	Distributed Training
	Word Embedding & Neural Networks	RNN LSTM	$\boxed{\text{Attention \& Transformers}}$	Generative Models & LLMs	Interpretability QA & RAG …

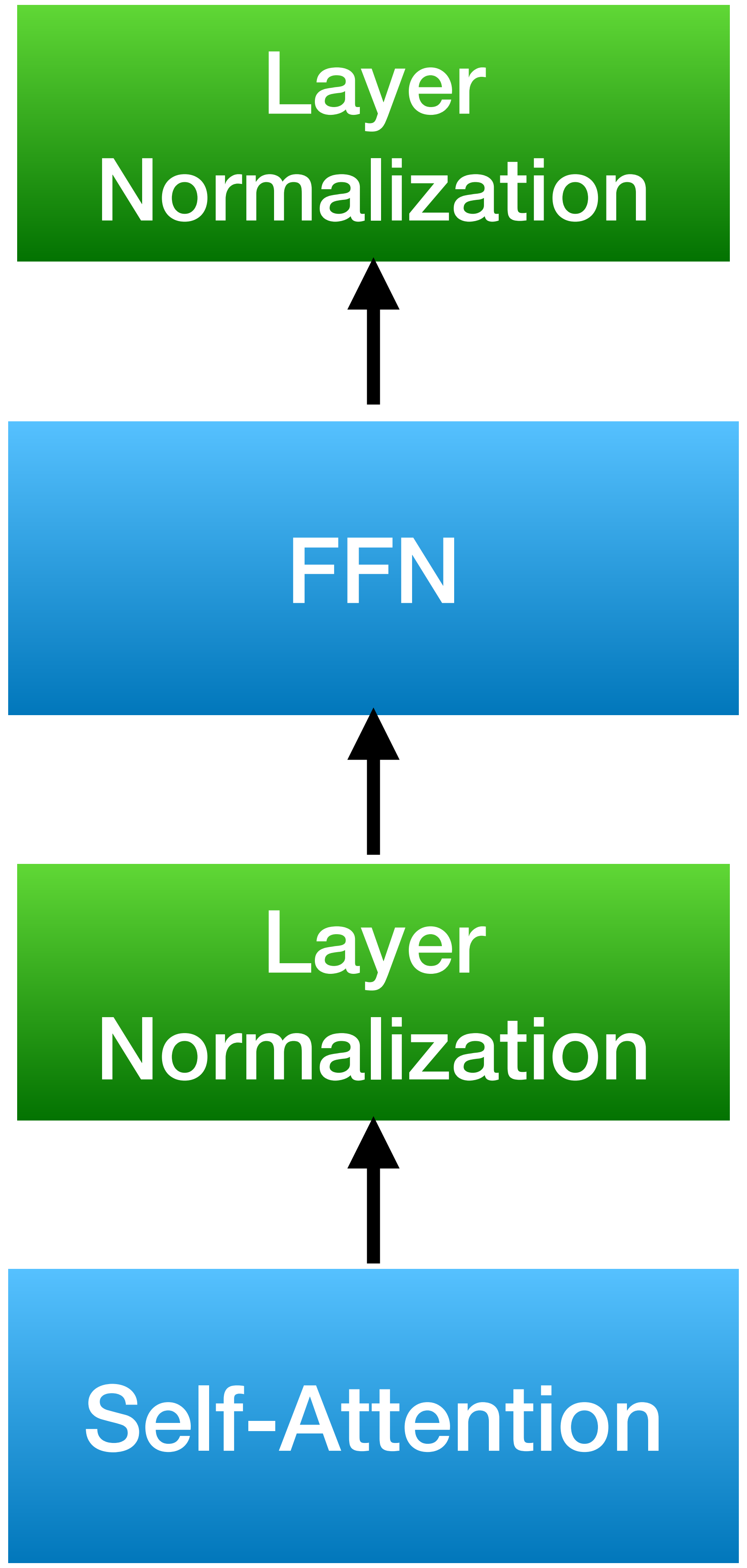
Transformers

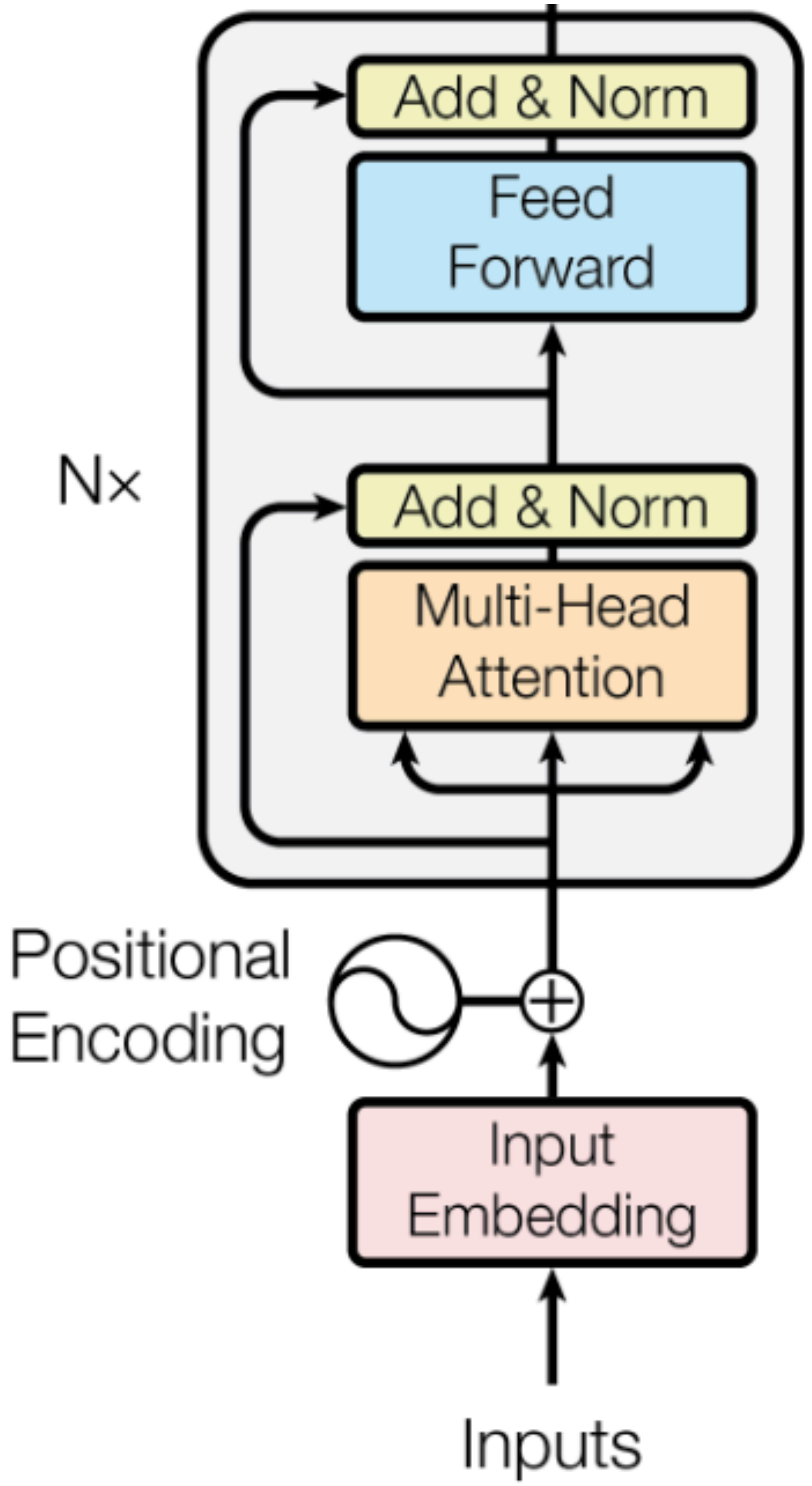
Transformer Encoder Block



- **Three Key Components**
  - (Masked) Multi-head Self-Attention
  - Layer Normalization
  - Position-wise Feed-Forward Network

Transformers





Outline

- **Residual Connections**
- **Normalization Layers**
  - Variants of Normalization Layers
  - Pre-Norm vs. Post-Norm
- **Positional Embeddings**
  - Absolute Positional Embeddings
  - Rotary Positional Embeddings
- **Other Variant Components**
  - FFN -> SwiGLU
  - MQA & GQA

Residual Connections

Residual Connections

Deep Residual Learning for Image Recognition

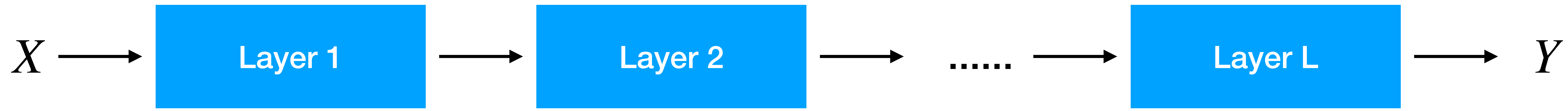
Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun
Microsoft Research
{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

Deep residual learning for **image recognition**
K He, X Zhang, S Ren, J Sun - … and **pattern recognition**, 2016 - openaccess.thecvf.com
… **Deeper** neural **networks** are more difficult to train. We present a **residual learning** framework to ease the training of **networks** that are substantially **deeper** than those used previously. …

Residual Connections

- **Deep Neural Networks**

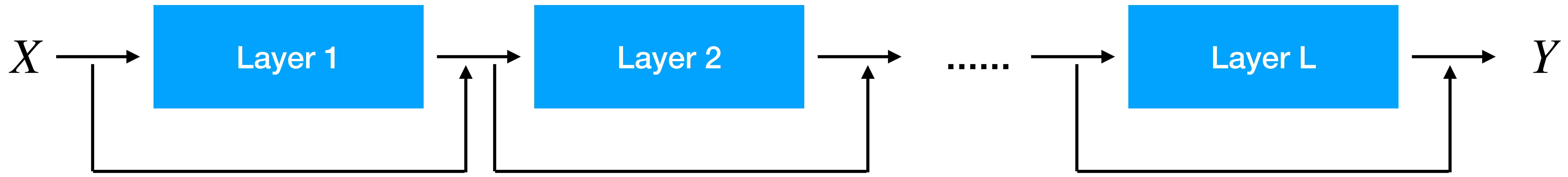
$$Y_1 = f_1(X) \qquad\qquad Y_2 = f_2(Y_1)$$



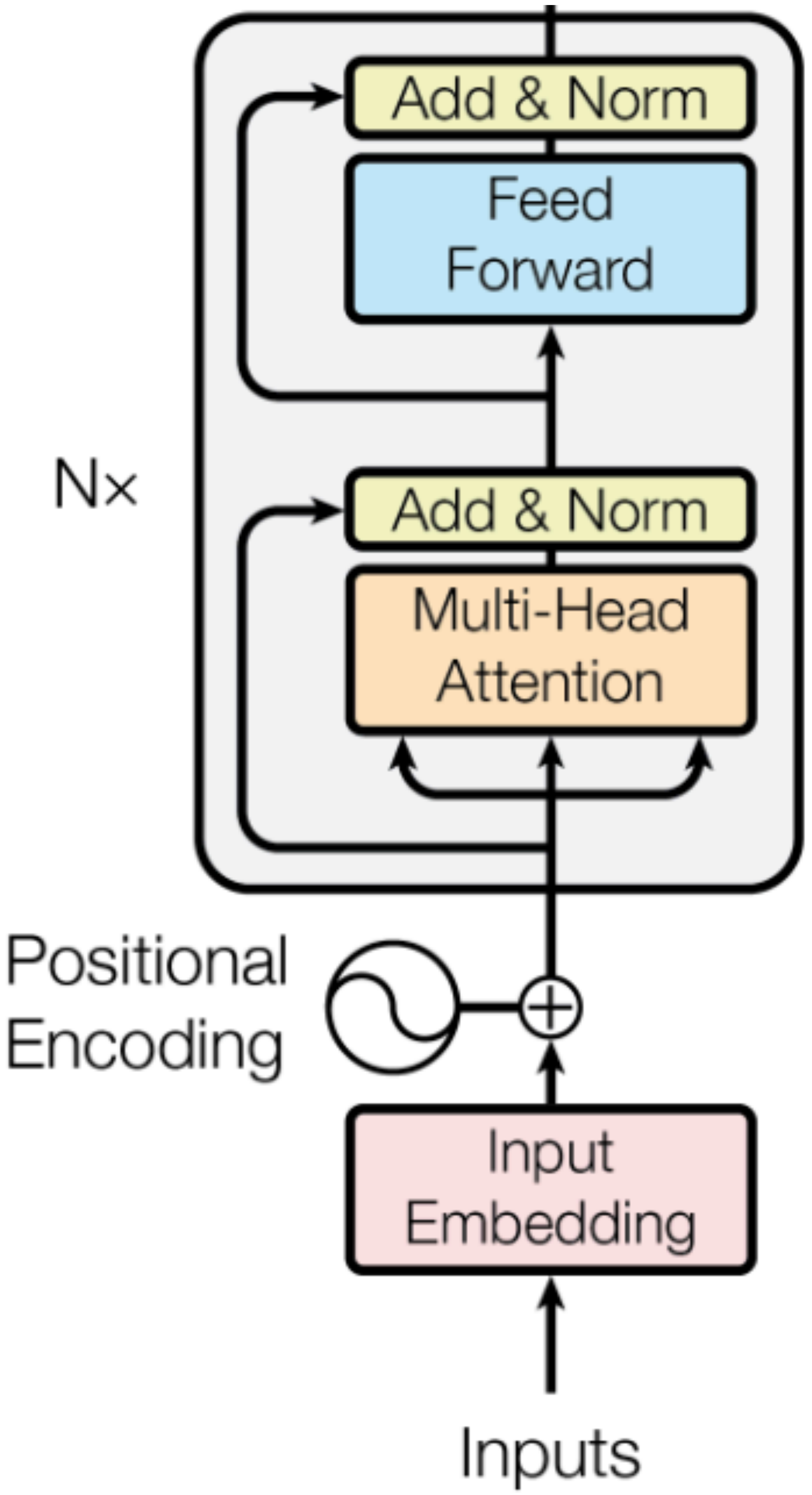
- **Residual Connections**

Does NOT change model capacity
but gradient flow in backpropogation

$$Y = f_1(X) + X \qquad\qquad Y_2 = f_2(Y_1) + Y_1$$



Residual Connection in Transformers



$$Y_1 = \textbf{LayerNorm}\left(\textbf{SelfAttn}(X) \boxed{+ X}\right)$$

$$Y_2 = \textbf{LayerNorm}\left(\textbf{FFN}(Y_1) \boxed{+ Y_1}\right)$$

Outline

- Residual Connections
- **Normalization Layers**
  - Variants of Normalization Layers
  - Pre-Norm vs. Post-Norm
- **Positional Embeddings**
  - Absolute Positional Embeddings
  - Rotary Positional Embeddings
- **Other Variant Components**
  - FFN -> SwiGLU
  - MQA & GQA

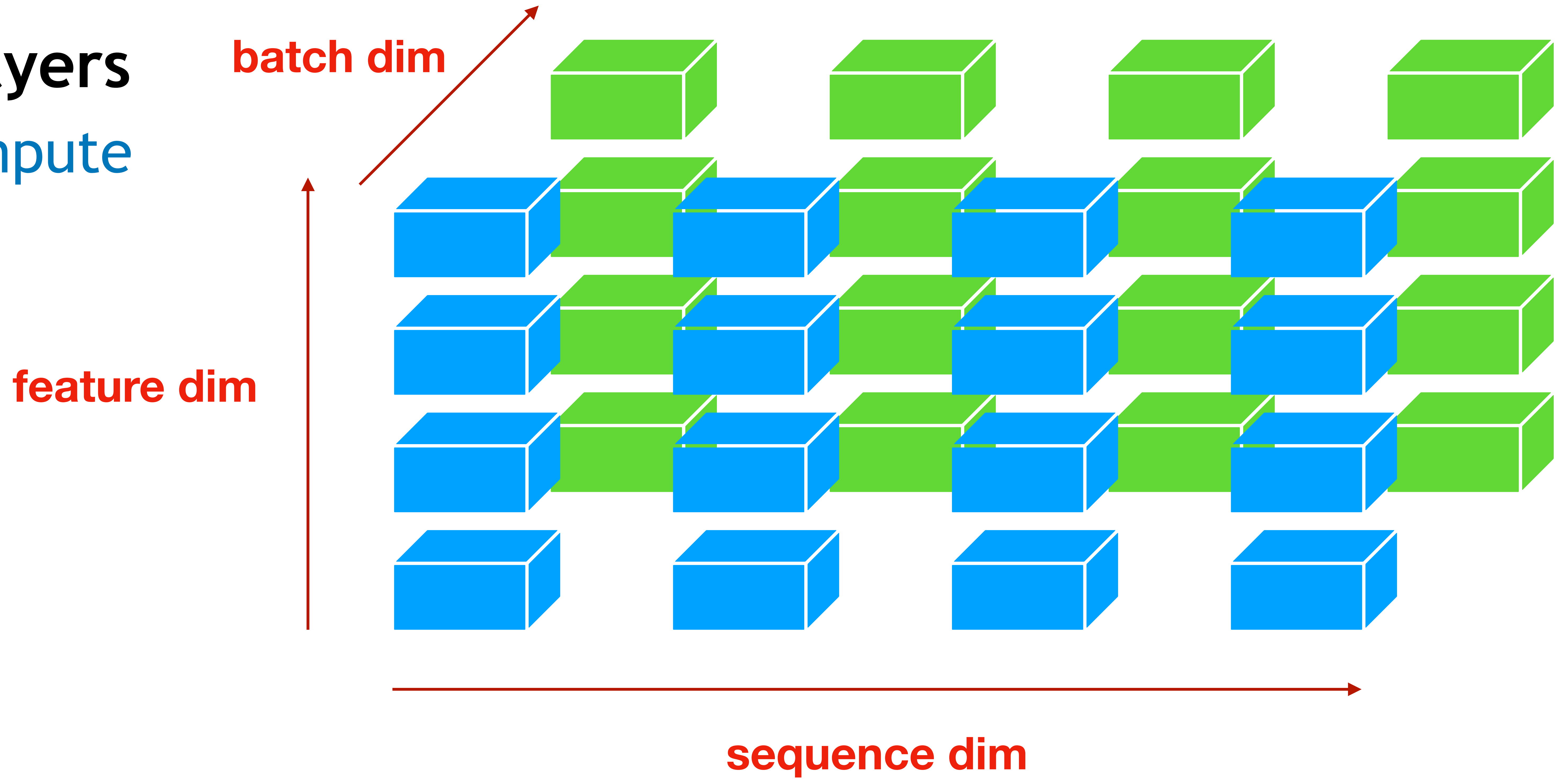
Normalization Layers

Normalization Layers

- **General Form:**

$$Y = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X] + \epsilon}} * \gamma + \beta$$

- **Variants of Normalization Layers**
  - Along which dimensions to compute expectation and variance
  - Batch Normalization
  - Layer Normalization
  - ……



Batch Normalization

- **First Proposed for CV models**
  – Loffe and Szegedy, 2014

Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

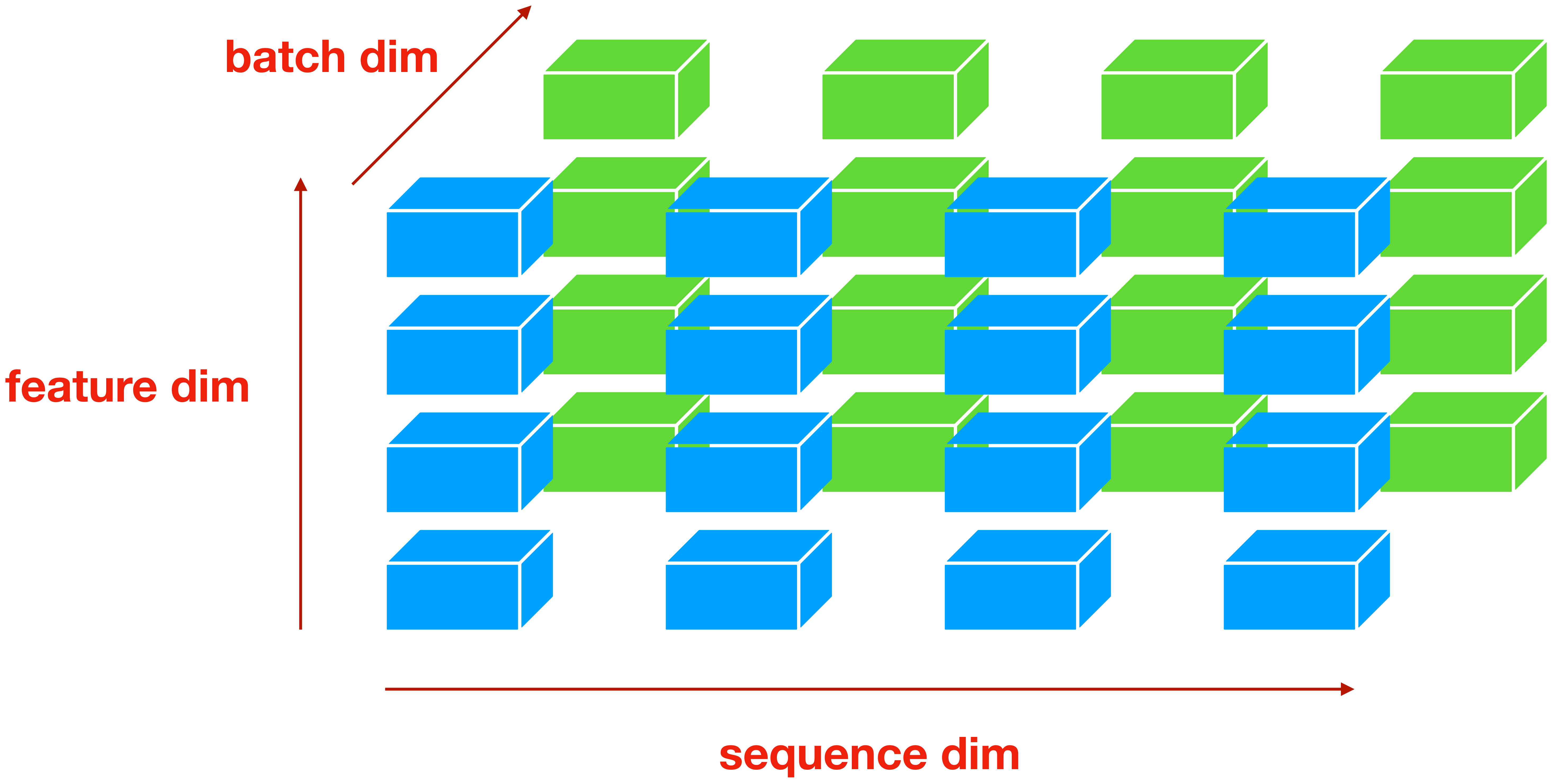
Sergey Ioffe SIOFFE@GOOGLE.COM
Christian Szegedy SZEGEDY@GOOGLE.COM
Google, 1600 Amphitheatre Pkwy, Mountain View, CA 94043

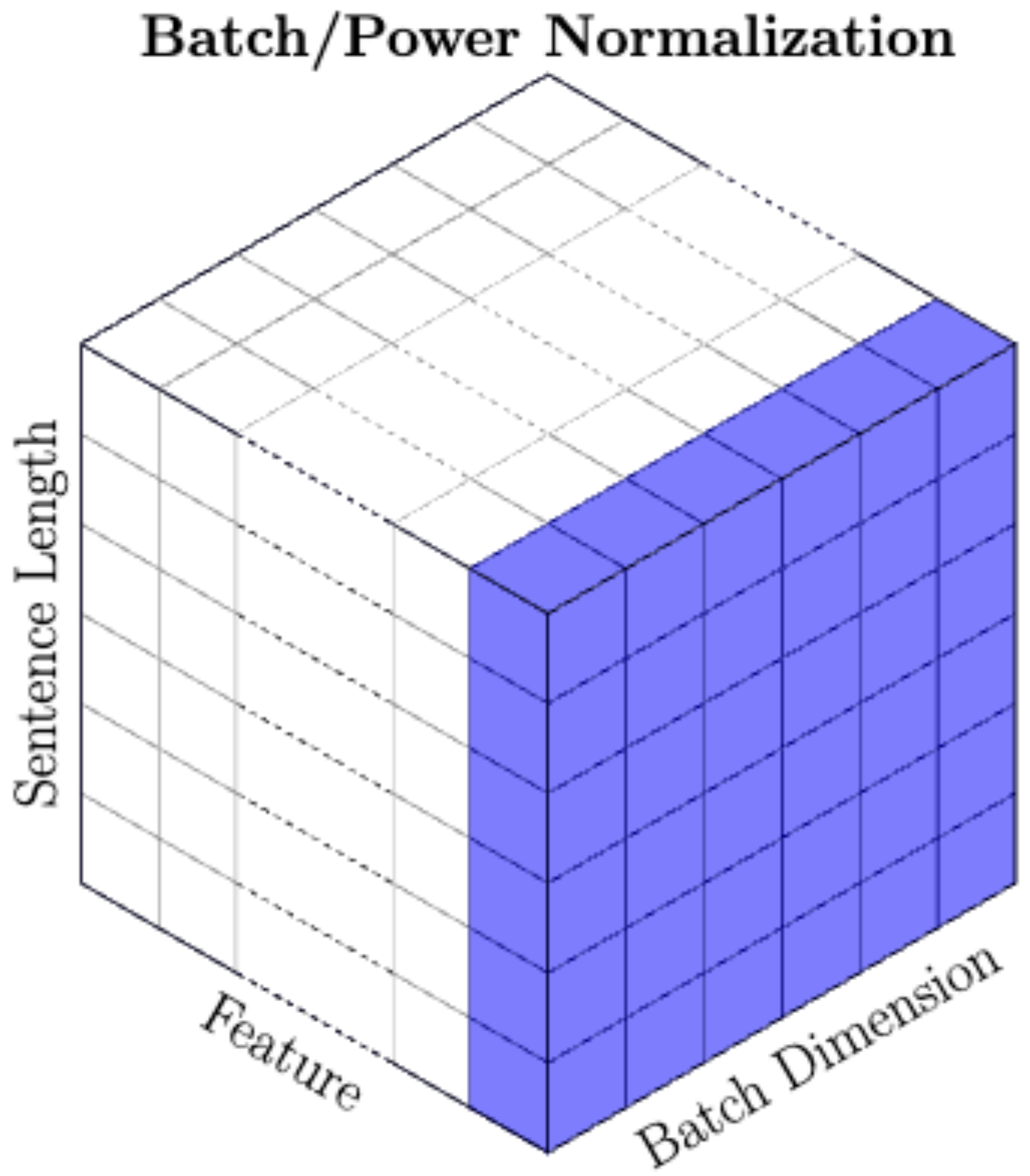
- **Two Basic Motivations**
  - Reducing internal covariate shift (bias) from data
  - Controlling output range/scale of each layer

Batch Normalization

- Normalize along both batch & spatial/sequential dimensions

$$Y = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X] + \epsilon}} * \gamma + \beta$$





Issues of Batch Normalization

- **Normalization involves batch dimension**
  - Batch size cannot be too small
  - When batch size < 8, batch normalization works poorly in practice

Hard to scale up

- **What are the expectation/variance in test time?**
  - Running stats for expectation/variance
  - At each mini-batch in training:
    - Compute mean and variance in this mini-batch

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m} x_i \qquad \text{// mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_{\mathcal{B}})^2 \qquad \text{// mini-batch variance}$$

    - Update running mean and variance

Brittle for data from different domains

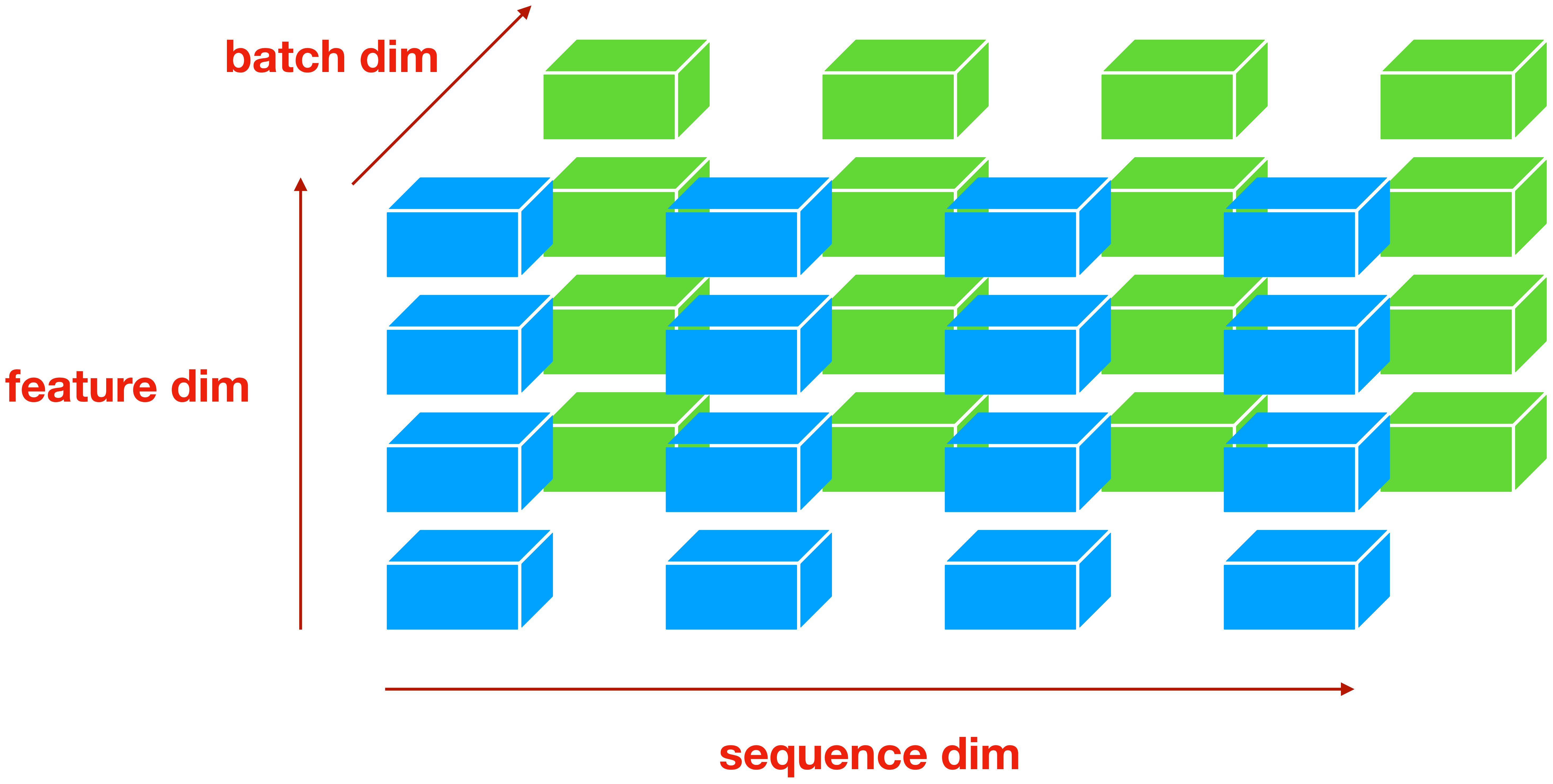
$$\mu \leftarrow \alpha\mu + (1-\alpha)\mu_{\mathcal{B}}$$

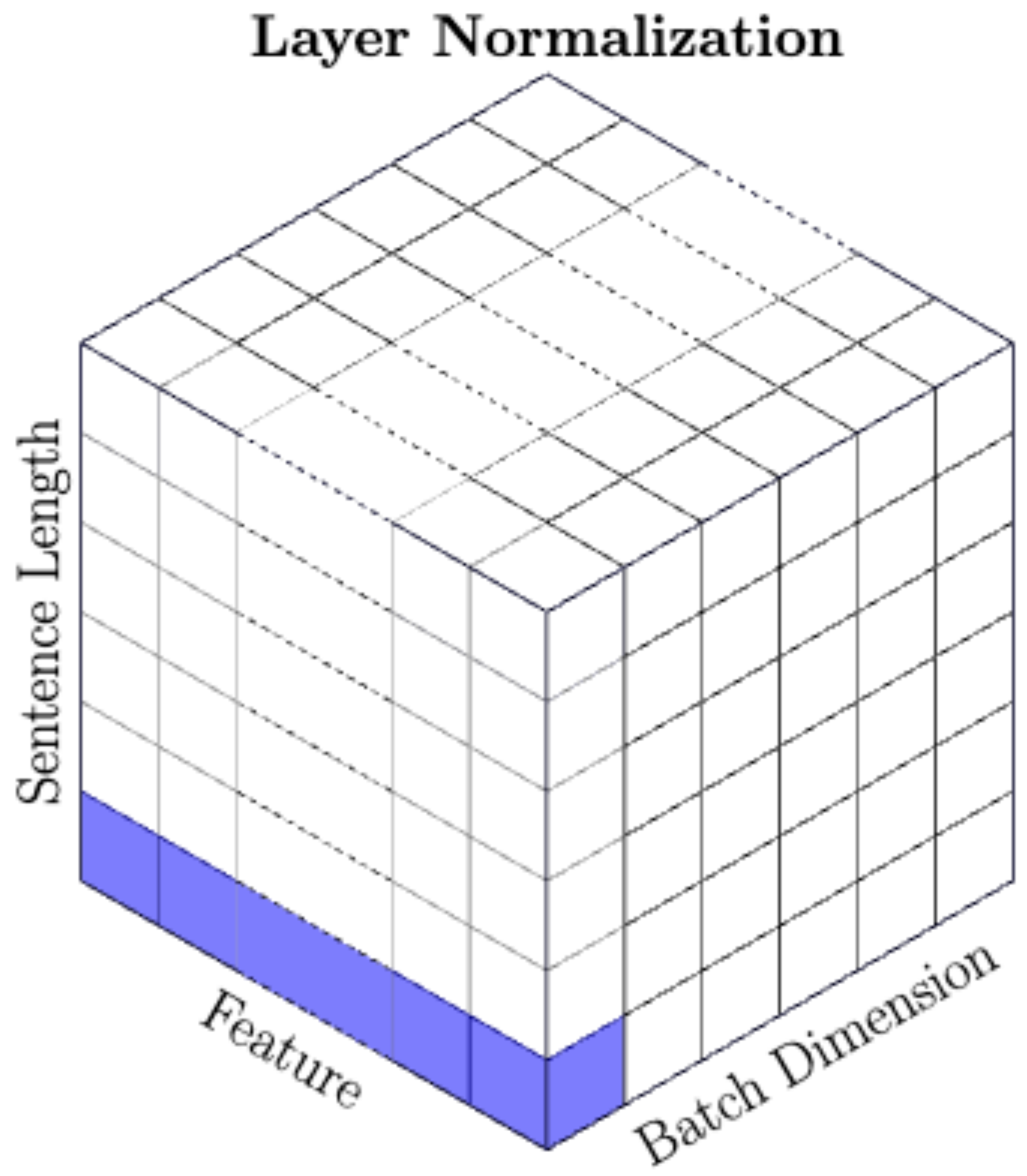
$$\sigma^2 \leftarrow \alpha\sigma^2 + (1-\alpha)\sigma_{\mathcal{B}}^2$$

Layer Normalization

- Normalize along the **feature** dimension

$$Y = \frac{X - \mathrm{E}[X]}{\sqrt{\mathrm{Var}[X] + \epsilon}} * \gamma + \beta$$





Layer Normalization

- **Pros:**
  - Does not involve the batch dimension (no need for running stats)
  - Does not involve sequential dimension
    - Easy to parallel
    - Working well for data w. various lengths (language)
- **Cons:**
  - Cannot reduce internal covariate shift/bias from data
    - Bias of data usually raises across different instances or steps
  - Root Mean Square (RMS) Normalization
    - Used in LLama

$$Y = \frac{X}{\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2 + \epsilon}} * \gamma$$

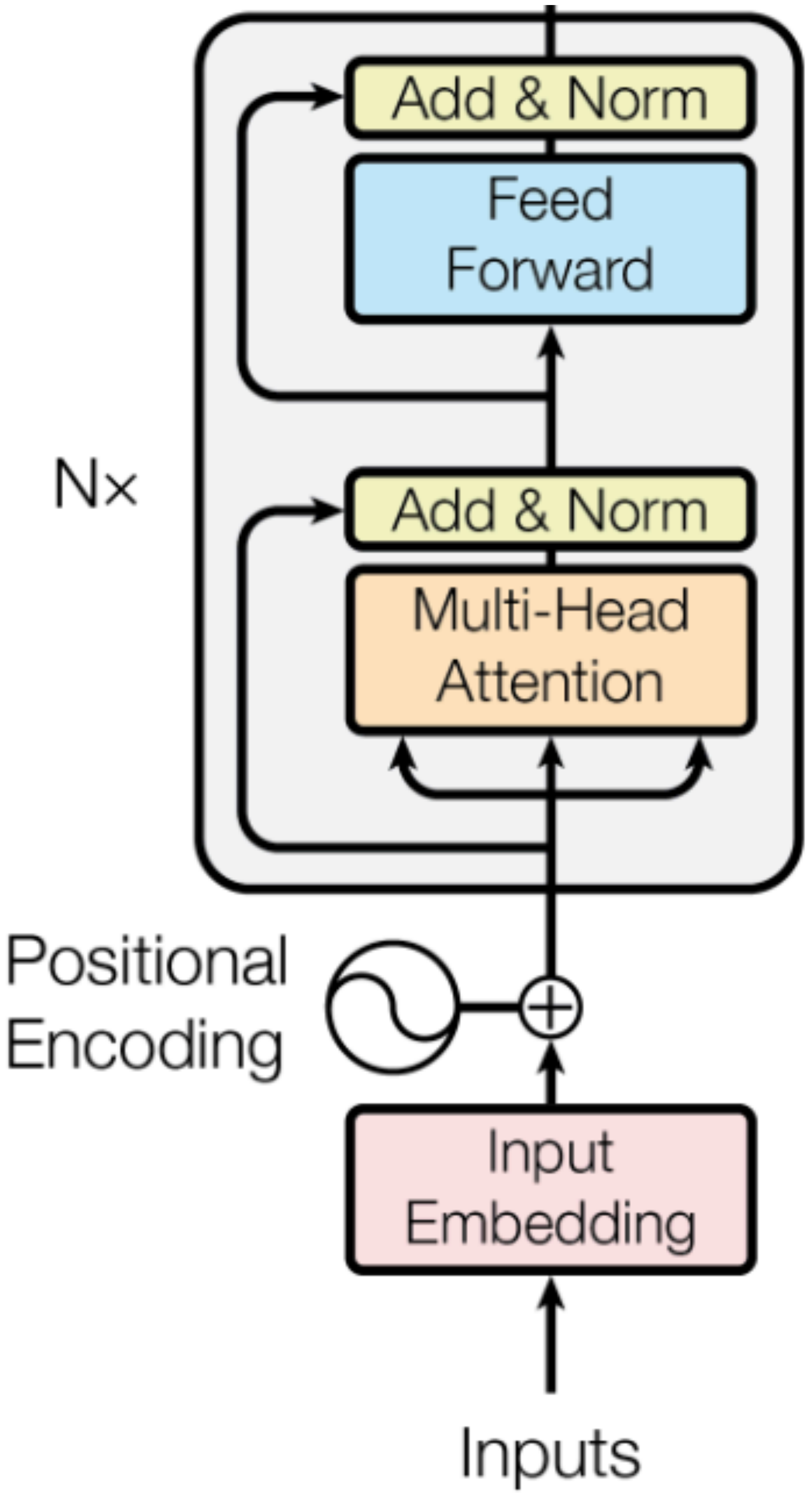
Outline

- Residual Connections
- **Normalization Layers**
  - Variants of Normalization Layers
  - Pre-Norm vs. Post-Norm
- **Positional Embeddings**
  - Absolute Positional Embeddings
  - Rotary Positional Embeddings
- **Other Variant Components**
  - FFN -> SwiGLU
  - MQA & GQA

Pre-Norm vs. Post-Norm

- **Position of Normalizations**
  - Normalization layers highly impact gradients in practice
  - The original Transformer architecture is hard to train w. more layers, even w. residual connections
  - Post-Norm -> Pre-Norm

Post-Normalization



$$Y_1 = \textbf{LayerNorm}\left(\textbf{SelfAttn}(X) \boxed{+X}\right)$$

$$Y_2 = \textbf{LayerNorm}\left(\textbf{FFN}(Y_1) \boxed{+Y_1}\right)$$

Detailed analysis

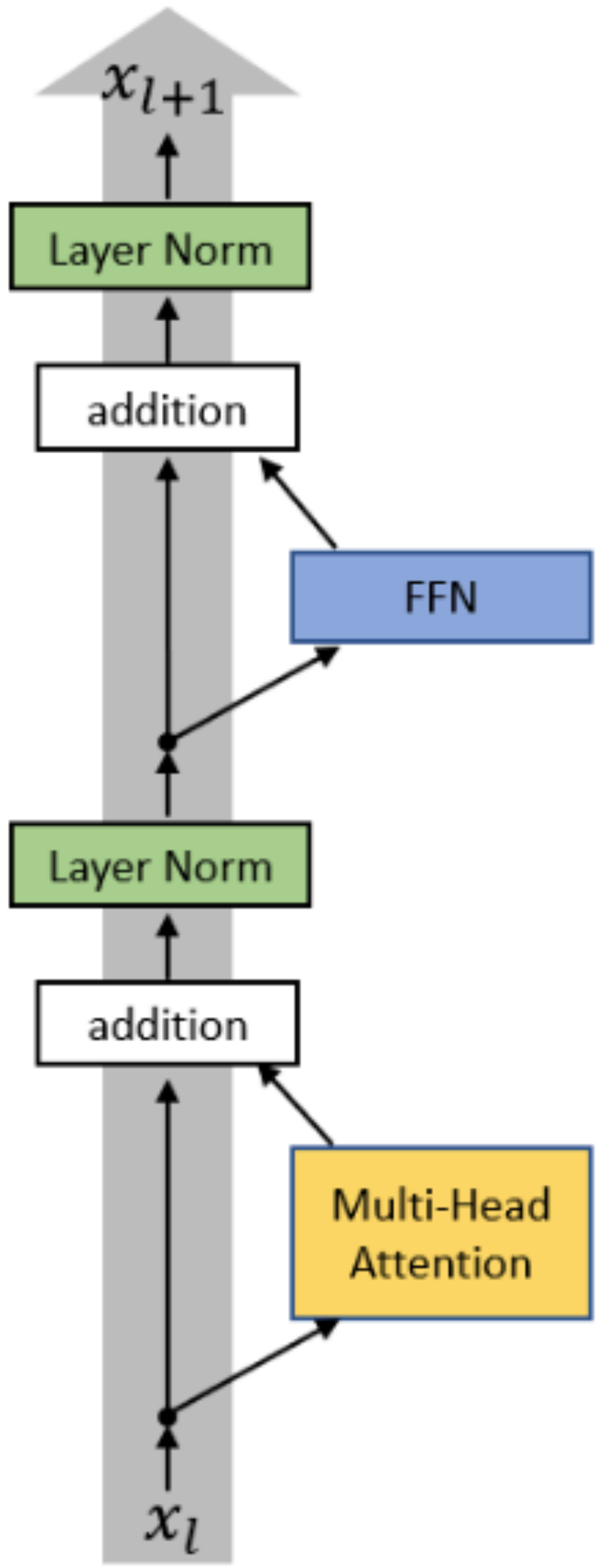
$$Y_1 = \textbf{LayerNorm}\left(f_1(X) + X\right)$$

$$Y_2 = \textbf{LayerNorm}\left(f_2(Y_1) + Y1\right) = \textbf{LayerNorm}\left(f_2(Y_1) + \textbf{LayerNorm}\left(f_1(X) + X\right)\right)$$

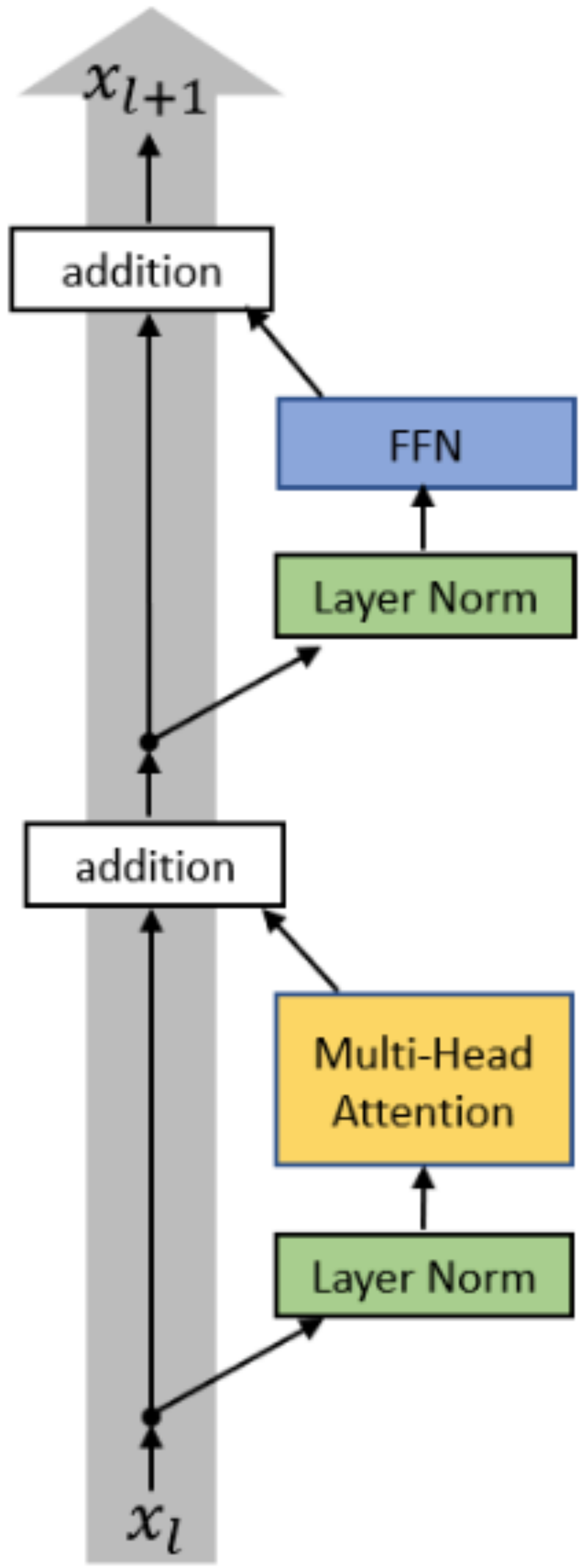
$$Y_l = \textbf{LayerNorm}\left(f_l(Y_{l-1}) + Y_{l-1}\right)$$

l layer-normalization blocks

Pre-Norm vs. Post-Norm

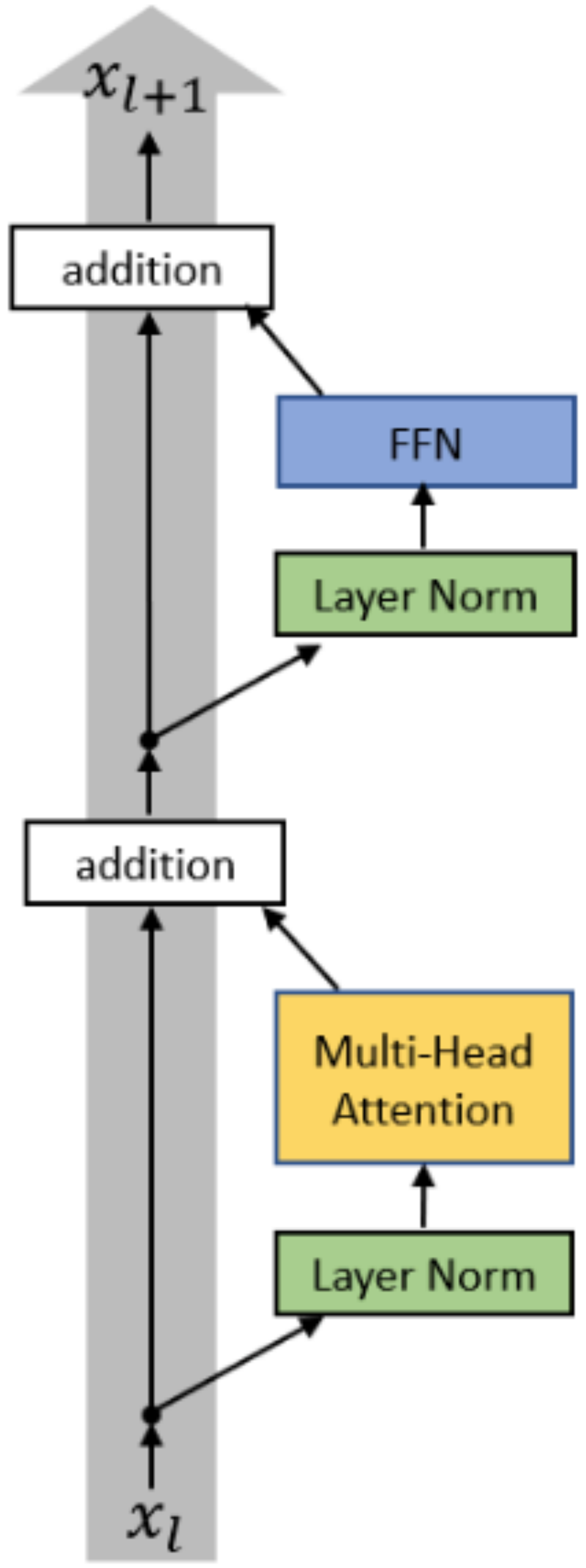


Post-Norm



Pre-Norm

Pre-Normalization



Pre-Norm

$$Y_1 = \textbf{SelfAttn}\left(\textbf{LayerNorm}(X)\right) \boxed{+ X}$$

$$Y_2 = \textbf{FFN}\left(\textbf{LayerNorm}(Y_1)\right) \boxed{+ Y_1}$$

Detailed analysis

$$Y_1 = f_1\left(\textbf{LayerNorm}(X)\right) + X$$

$$Y_2 = f_2\left(\textbf{LayerNorm}(Y_1)\right) + Y1 = f_2\left(\textbf{LayerNorm}(Y_1)\right) + f_1\left(\textbf{LayerNorm}(X)\right) + X$$

1 layer-normalization block for each layer output

Pre-Normalization

- **Pros:**
  - Keeping (almost) all good properties of post-norm
  - Similar performance w. Post-norm
  - Able to train very deep Transformer models
- **Cons:**
  - Numerically unstable (training spikes)

What is the range of Y_2 in post-norm and pre-norm?

Post-norm $$Y_2 = \textbf{LayerNorm}\left(f_2(Y_1) + Y1\right) = \textbf{LayerNorm}\left(f_2(Y_1) + \textbf{LayerNorm}\left(f_1(X) + X\right)\right)$$

Pre-norm $$Y_2 = f_2\left(\textbf{LayerNorm}(Y_1)\right) + Y1 = f_2\left(\textbf{LayerNorm}(Y_1)\right) + f_1\left(\textbf{LayerNorm}(X)\right) + X$$

Outline

- **Residual Connections**
- **Normalization Layers**
  - Variants of Normalization Layers
  - Pre-Norm vs. Post-Norm
- **Positional Embeddings**
  - Absolute Positional Embeddings
  - Rotary Positional Embeddings
- **Other Variant Components**
  - FFN -> SwiGLU
  - MQA & GQA

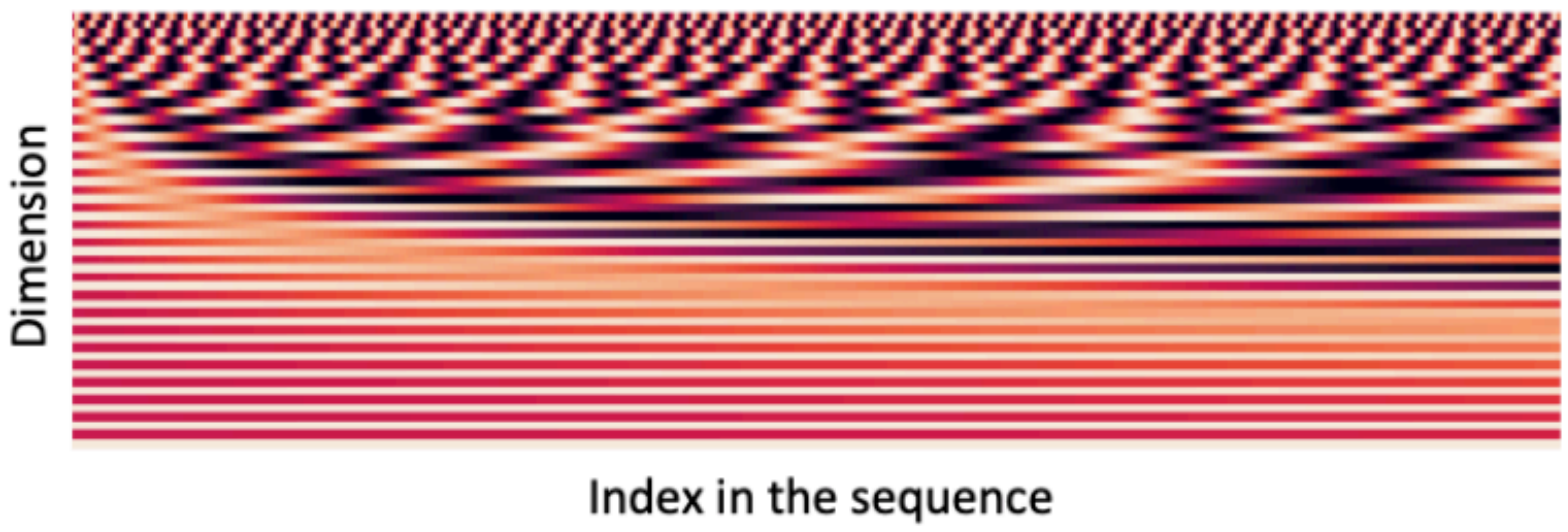
Positional Information

- **Unlike RNNs, self-attention does not build in order information**
  - Encode the order of the sentence into the input $x_1, \ldots, x_n$
- **Solution: add positional encoding to the input embeddings**

$$x_i \leftarrow x_i + p_i$$

- **Use sine and cosine functions of different frequencies (not learnable)**

$$p_i = \begin{pmatrix} \sin(i/10000^{2*1/d}) \\ \cos(i/10000^{2*1/d}) \\ \vdots \\ \sin(i/10000^{2*\frac{d}{2}/d}) \\ \cos(i/10000^{2*\frac{d}{2}/d}) \end{pmatrix}$$



Sinusoidal Positional Embedding

$$p_{t,j} = \sin\left(t \cdot \theta^{\frac{j}{2d}}\right) \quad j \,\%\, 2 = 0$$

$$p_{t,j} = \cos\left(t \cdot \theta^{\frac{j-1}{2d}}\right) \quad j \,\%\, 2 = 1$$

$$\theta = \frac{1}{10000}$$

Position $t = 0,1,\ldots,n$

dim $j = 0,\ldots,d$

	t=0	t=1	t=2	…	t=n
j=0	$\sin(0 \cdot \theta^0)$	$\sin(1 \cdot \theta^0)$	$\sin(2 \cdot \theta^0)$		$\sin(n \cdot \theta^0)$
j=1	$\cos(0 \cdot \theta^0)$	$\cos(1 \cdot \theta^0)$	$\cos(2 \cdot \theta^0)$		$\cos(n \cdot \theta^0)$
j=2	$\sin(0 \cdot \theta^{\frac{1}{2d}})$	$\sin(1 \cdot \theta^{\frac{1}{2d}})$	$\sin(2 \cdot \theta^{\frac{1}{2d}})$		$\sin(n \cdot \theta^{\frac{1}{2d}})$
j=3	$\cos(0 \cdot \theta^{\frac{1}{2d}})$	$\cos(1 \cdot \theta^{\frac{1}{2d}})$	$\cos(2 \cdot \theta^{\frac{1}{2d}})$		$\cos(n \cdot \theta^{\frac{1}{2d}})$

Sinusoidal Positional Embedding

$$p_{t,j} = \sin\left(t \cdot \theta^{\frac{j}{2d}}\right) \quad j \,\%\, 2 = 0$$

$$p_{t,j} = \cos\left(t \cdot \theta^{\frac{j-1}{2d}}\right) \quad j \,\%\, 2 = 1$$

$$\theta = \frac{1}{10000}$$

Position $t = 0,1,\ldots,n$

dim $j = 0,\ldots,d$

	t=0	t=1	t=2	…	t=n
j=0	$\sin(0 \cdot \theta^0)$	$\sin(1 \cdot \theta^0)$	$\sin(2 \cdot \theta^0)$		$\sin(n \cdot \theta^0)$
j=1	$\cos(0 \cdot \theta^0)$	$\cos(1 \cdot \theta^0)$	$\cos(2 \cdot \theta^0)$		$\cos(n \cdot \theta^0)$
j=2	$\sin(0 \cdot \theta^{\frac{1}{2d}})$	$\sin(1 \cdot \theta^{\frac{1}{2d}})$	$\sin(2 \cdot \theta^{\frac{1}{2d}})$		$\sin(n \cdot \theta^{\frac{1}{2d}})$
j=3	$\cos(0 \cdot \theta^{\frac{1}{2d}})$	$\cos(1 \cdot \theta^{\frac{1}{2d}})$	$\cos(2 \cdot \theta^{\frac{1}{2d}})$		$\cos(n \cdot \theta^{\frac{1}{2d}})$

Sinusoidal Positional Embedding

$$p_{t,j} = \sin\left(t \cdot \theta^{\frac{j}{2d}}\right) \qquad j \,\%\, 2 = 0$$

$$p_{t,j} = \cos\left(t \cdot \theta^{\frac{j-1}{2d}}\right) \qquad j \,\%\, 2 = 1$$

$$\theta = \frac{1}{10000}$$

Position $t = 0,1,\ldots,n$

dim $j = 0,\ldots,d$

	t=0	t=1	t=2	…	t=n
j=0	$\sin(0 \cdot \theta^{0})$	$\sin(1 \cdot \theta^{0})$	$\sin(2 \cdot \theta^{0})$		$\sin(n \cdot \theta^{0})$
j=1	$\cos(0 \cdot \theta^{0})$	$\cos(1 \cdot \theta^{0})$	$\cos(2 \cdot \theta^{0})$		$\cos(n \cdot \theta^{0})$
j=2	$\sin(0 \cdot \theta^{\frac{1}{2d}})$	$\sin(1 \cdot \theta^{\frac{1}{2d}})$	$\sin(2 \cdot \theta^{\frac{1}{2d}})$		$\sin(n \cdot \theta^{\frac{1}{2d}})$
j=3	$\cos(0 \cdot \theta^{\frac{1}{2d}})$	$\cos(1 \cdot \theta^{\frac{1}{2d}})$	$\cos(2 \cdot \theta^{\frac{1}{2d}})$		$\cos(n \cdot \theta^{\frac{1}{2d}})$

Sinusoidal Positional Embedding

$$p_{t,j} = \sin\left(t \cdot \theta^{\frac{j}{2d}}\right) \qquad j \,\%\, 2 = 0$$

$$p_{t,j} = \cos\left(t \cdot \theta^{\frac{j-1}{2d}}\right) \qquad j \,\%\, 2 = 1$$

$$\theta = \frac{1}{10000}$$

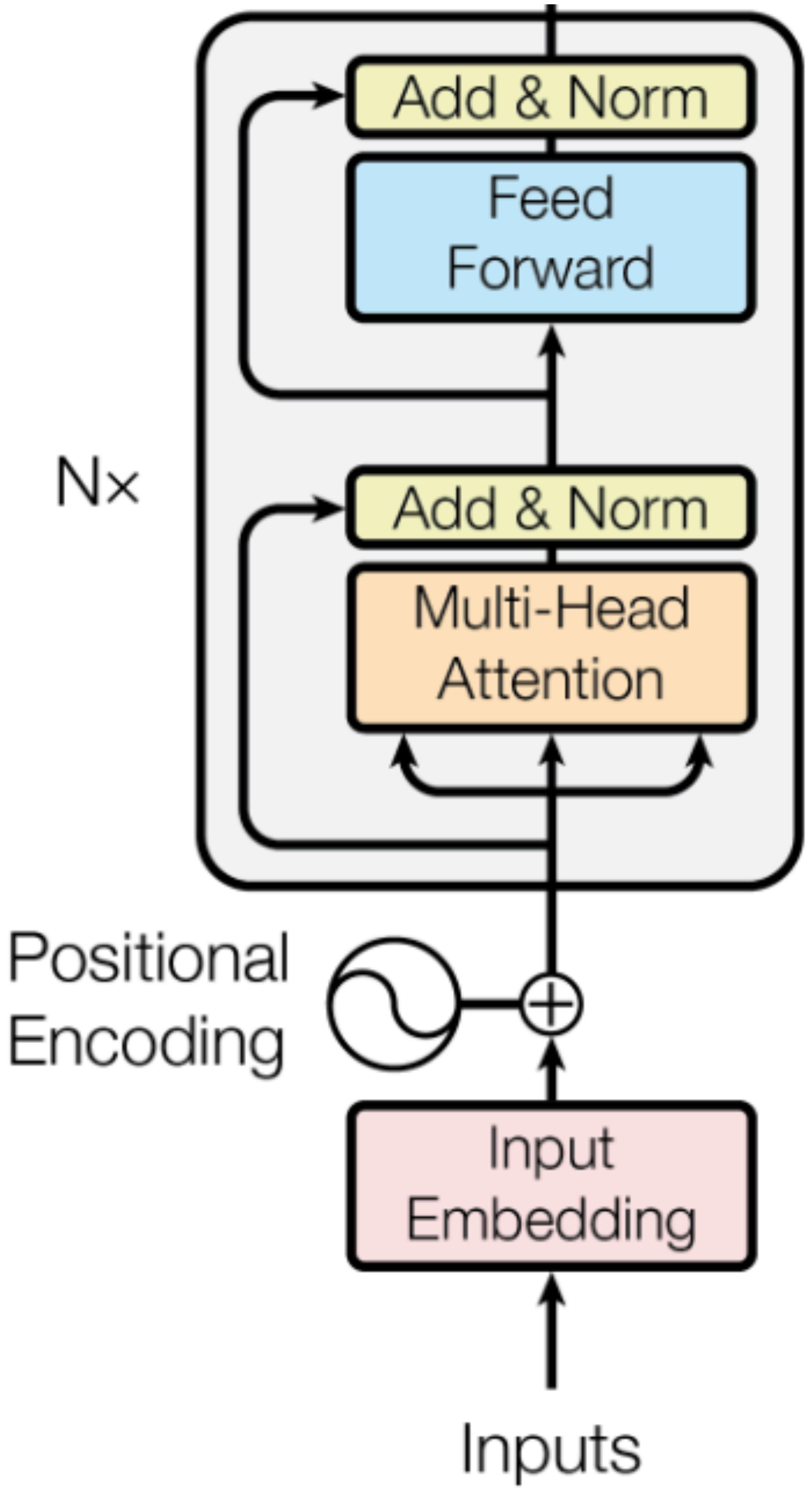
Position $t = 0,1,\ldots,n$

dim $j = 0,\ldots,d$

	t=0	t=1	t=2	…	t=n
j=0	$\sin(0 \cdot \theta^0)$	$\sin(1 \cdot \theta^0)$	$\sin(2 \cdot \theta^0)$		$\sin(n \cdot \theta^0)$
j=1	$\cos(0 \cdot \theta^0)$	$\cos(1 \cdot \theta^0)$	$\cos(2 \cdot \theta^0)$		$\cos(n \cdot \theta^0)$
j=2	$\sin(0 \cdot \theta^{\frac{1}{2d}})$	$\sin(1 \cdot \theta^{\frac{1}{2d}})$	$\sin(2 \cdot \theta^{\frac{1}{2d}})$		$\sin(n \cdot \theta^{\frac{1}{2d}})$
j=3	$\cos(0 \cdot \theta^{\frac{1}{2d}})$	$\cos(1 \cdot \theta^{\frac{1}{2d}})$	$\cos(2 \cdot \theta^{\frac{1}{2d}})$		$\cos(n \cdot \theta^{\frac{1}{2d}})$

Sinusoidal Positional Embedding

- **Sinusoidal positional embedding depends on absolute positional information**
- **Only added to the input word embeddings**
- **Ability for length extension**
  - Training on up to k words
  - Test on > k words



Rotary Positional Embedding

- **Absolute positional index is not meaningful**
  - I don’t know
  - In fact, I don’t know

			I	**don’t**	**know**
			p_1	p_2	p_3
In	**fact**	**,**	**I**	**don’t**	**know**
p_1	p_2	p_3	p_4	p_5	p_6

Rotary Positional Embedding

- **Absolute positional index is not meaningful**
  - I don't know
  - In fact, I don't know
- **How about relative positional embedding?**
  - Positional information should be invariant w.r.t relative distance of two positions
  - Rotary Position Embedding
    - Used in Llama models

Rotary Positional Embedding

Rotary Operation

$$X \in \mathbb{R}^d \qquad \theta_j = \left(\frac{1}{10000}\right)^{\frac{j}{2d}}$$

$$\textbf{Rotary}(X, m) = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_d \\ x_{d-1} \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix}$$

Rotary Positional Embedding

Attention

$$\boxed{\mathrm{attn}(Q, K, V) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d}})V}$$

Applying rotary operation to each query and key vector

$$q_i' = \textbf{Rotary}(q_i, i)$$

$$k_j' = \textbf{Rotary}(k_j, j)$$

Replacing Q, K with Q', K' to compute attention

Rotary Positional Embedding

- **A simple case:** $d = 2$

$$\textbf{Rotary}(X, m) = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_d \\ x_{d-1} \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix}$$

$$q_i' = \mathrm{Rotary}(q_i, i) = \begin{pmatrix} q_{i,1}\cos(i\theta) - q_{i,2}\sin(i\theta) \\ q_{i,1}\sin(i\theta) + q_{i,2}\cos(i\theta) \end{pmatrix}$$

$$k_j' = \mathrm{Rotary}(k_j, j) = \begin{pmatrix} k_{j,1}\cos(j\theta) - k_{j,2}\sin(j\theta) \\ k_{j,1}\sin(j\theta) + k_{j,2}\cos(j\theta) \end{pmatrix}$$

Rotary Positional Embedding

- **A simple case:** $d = 2$

$$q_i' = \mathrm{Rotary}(q_i, i) = \begin{pmatrix} q_{i,1}\cos(i\theta) - q_{i,2}\sin(i\theta) \\ q_{i,1}\sin(i\theta) + q_{i,2}\cos(i\theta) \end{pmatrix}$$

$$k_j' = \mathrm{Rotary}(k_j, j) = \begin{pmatrix} k_{j,1}\cos(j\theta) - k_{j,2}\sin(j\theta) \\ k_{j,1}\sin(j\theta) + k_{j,2}\cos(j\theta) \end{pmatrix}$$

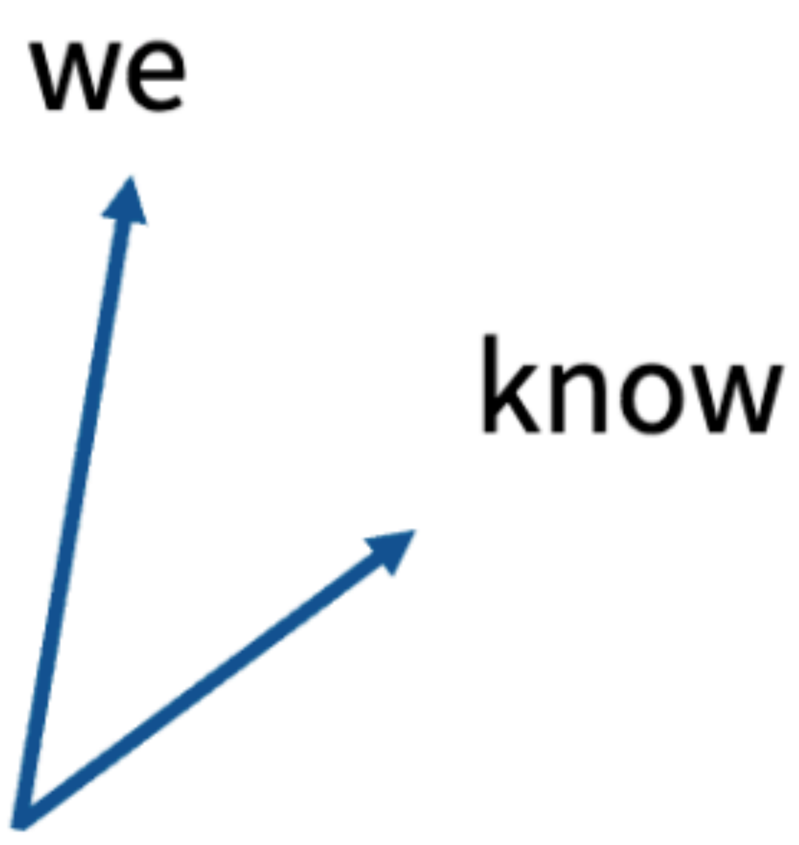
$$< q_i', k_j' > = (q_{i,1}\cos(i\theta) - q_{i,2}\sin(i\theta))(k_{j,1}\cos(j\theta) - k_{j,2}\sin(j\theta))$$

$$+(q_{i,1}\sin(i\theta) + q_{i,2}\cos(i\theta))(k_{j,1}\sin(j\theta) + k_{j,2}\cos(j\theta))$$

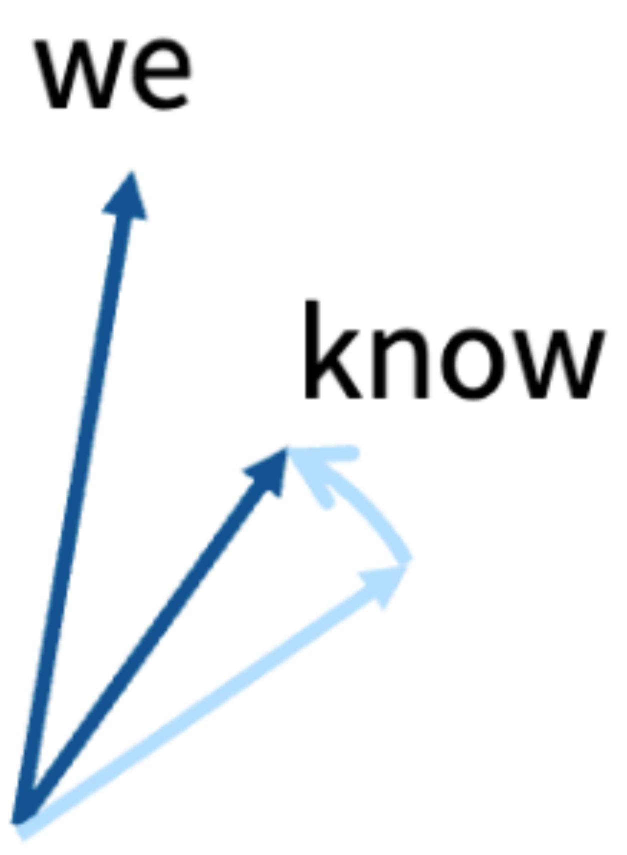
$$= \left(q_{i,1}k_{j,1} + q_{i,2}k_{j,2}\right)\cos\boxed{\left((i-j)\theta\right)} + \left(q_{i,1}k_{j,2} - q_{i,2}k_{j,1}\right)\sin\boxed{\left((i-j)\theta\right)}$$

$< q_i', k_j' >$ only depends on $(i-j)\theta$

Rotary Positional Embedding

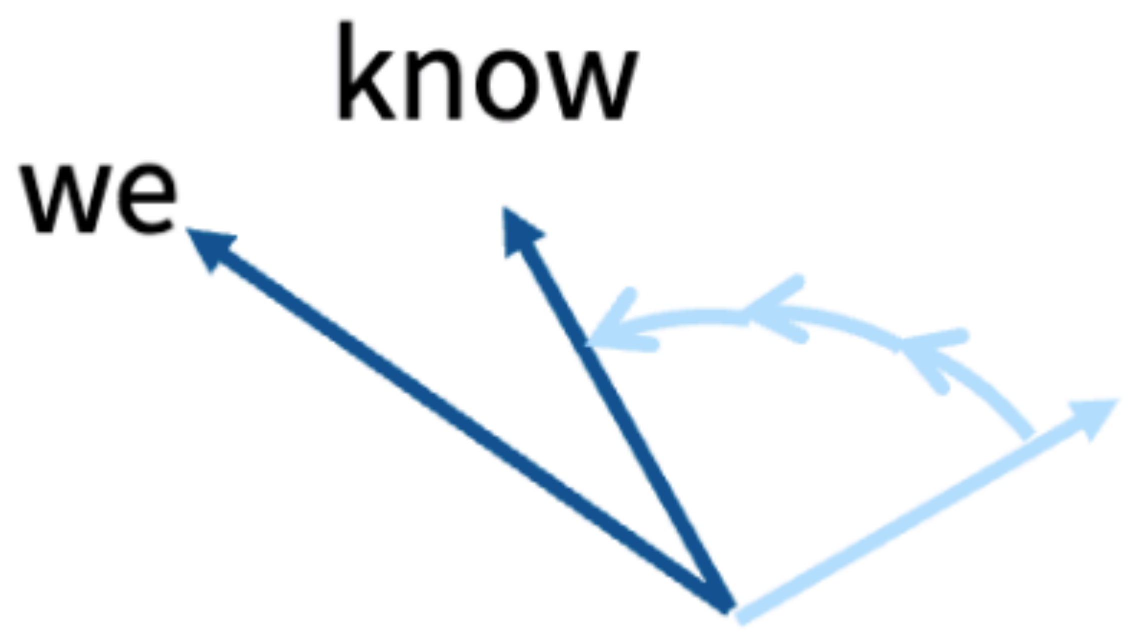


Position independent
embedding



Embedding
“we know that”

Rotate we by ‘0 positions’
know by ‘1 positions’



Embedding
“of course we know”

Rotate we by ‘2 positions’
Rotate know by ‘3 positions’

Outline

- **Residual Connections**
- **Normalization Layers**
  - Variants of Normalization Layers
  - Pre-Norm vs. Post-Norm
- **Positional Embeddings**
  - Absolute Positional Embeddings
  - Rotary Positional Embeddings
- **Other Variant Components**
  - FFN -> SwiGLU
  - MQA & GQA

SwiGLU

SwiGLU (swish is $x * \text{sigmoid}(x)$**)**

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(x, W, V, W_2) = (\mathrm{Swish}_1(xW) \otimes xV)W_2$$

Notable models:

LLaMa 1/2/3, PaLM, Mistral, OlMo, *most models post 2023*

Note: Gated models use smaller dimensions for the d_{ff} by 2/3

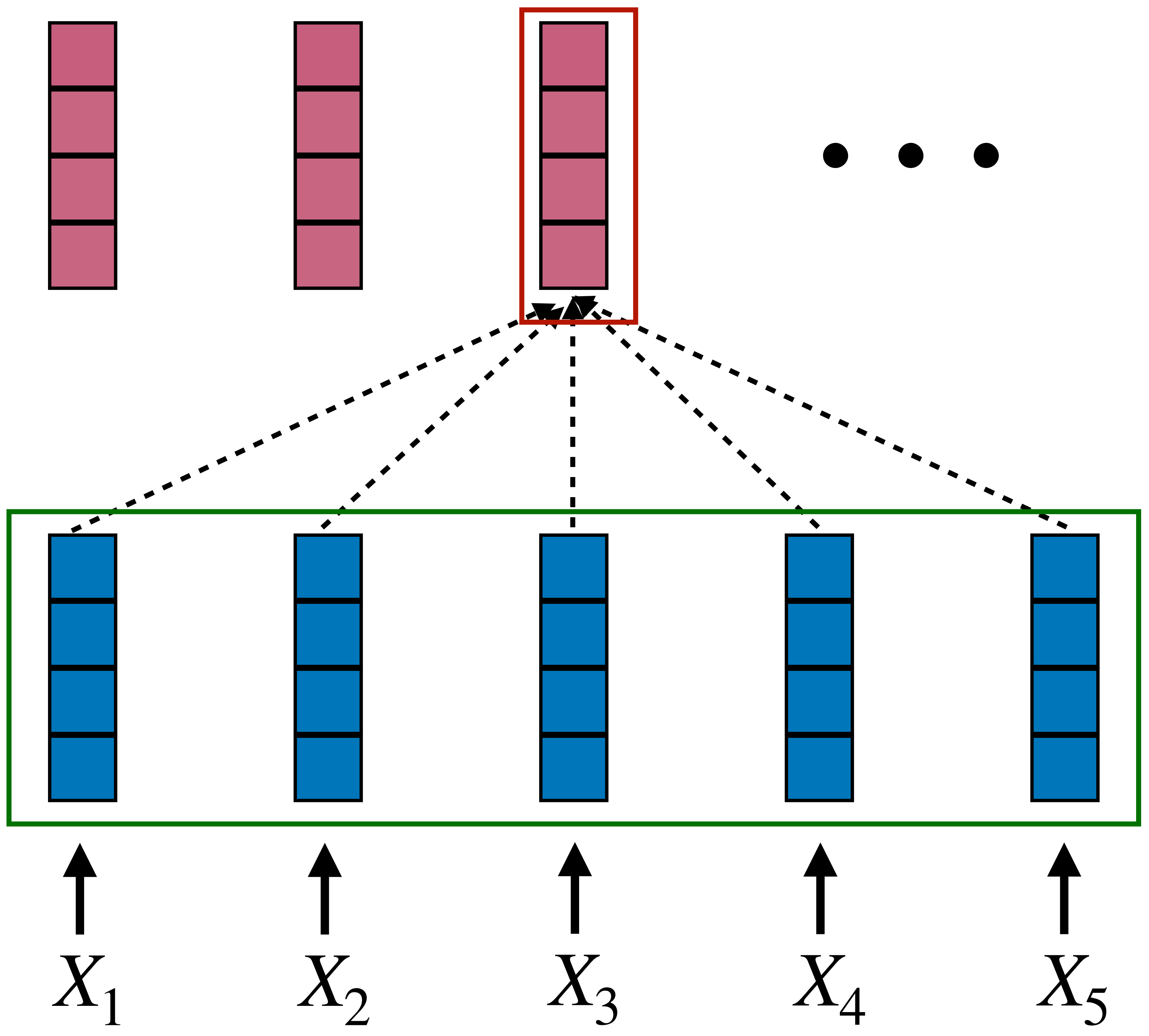
Do Gated Linear Units Work?

	Score Average	CoLA MCC	SST-2 Acc
FFN_{ReLU}	83.80	51.32	94.04
FFN_{GELU}	83.86	53.48	94.04
$\text{FFN}_{\text{Swish}}$	83.60	49.79	93.69
FFN_{GLU}	84.20	49.16	94.27
$\text{FFN}_{\text{GEGLU}}$	84.12	53.65	93.92
$\text{FFN}_{\text{Bilinear}}$	83.79	51.02	**94.38**
$\text{FFN}_{\text{SwiGLU}}$	84.36	51.59	93.92
$\text{FFN}_{\text{ReGLU}}$	**84.67**	**56.16**	**94.38**
[Raffel et al., 2019]	83.28	53.84	92.68
ibid. stddev.	0.235	1.111	0.569

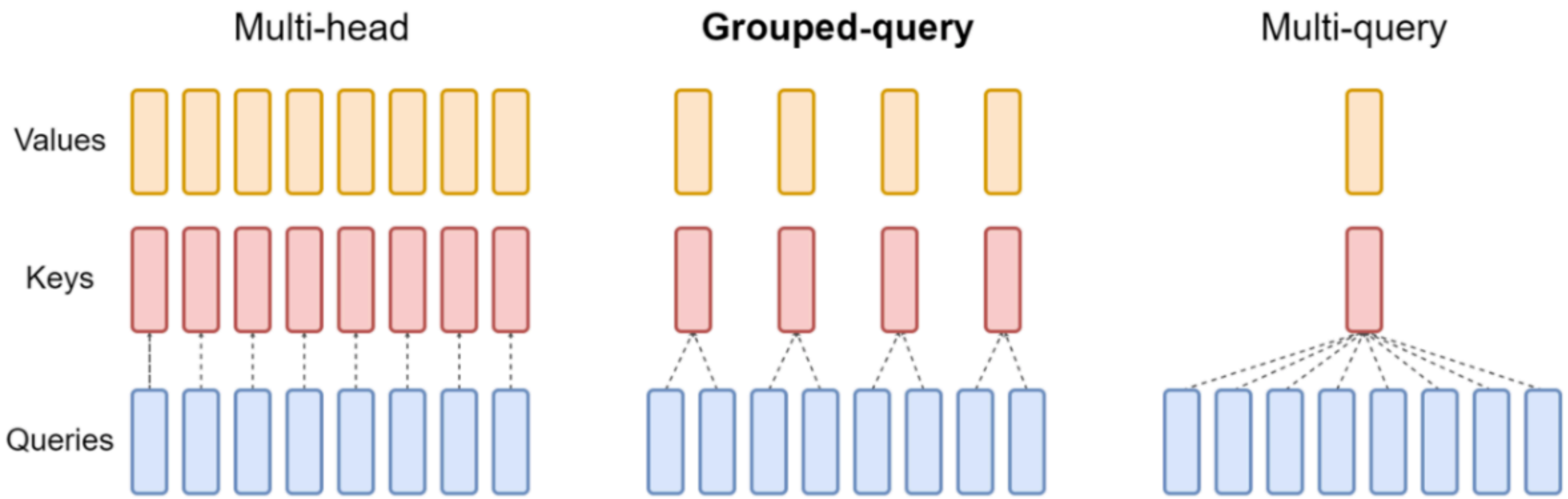
Shazeer 2020

MQA & GQA

- **KV cache in Transformer inference**



MQA & GQA



Do we also save computes w. GQA/MQA?

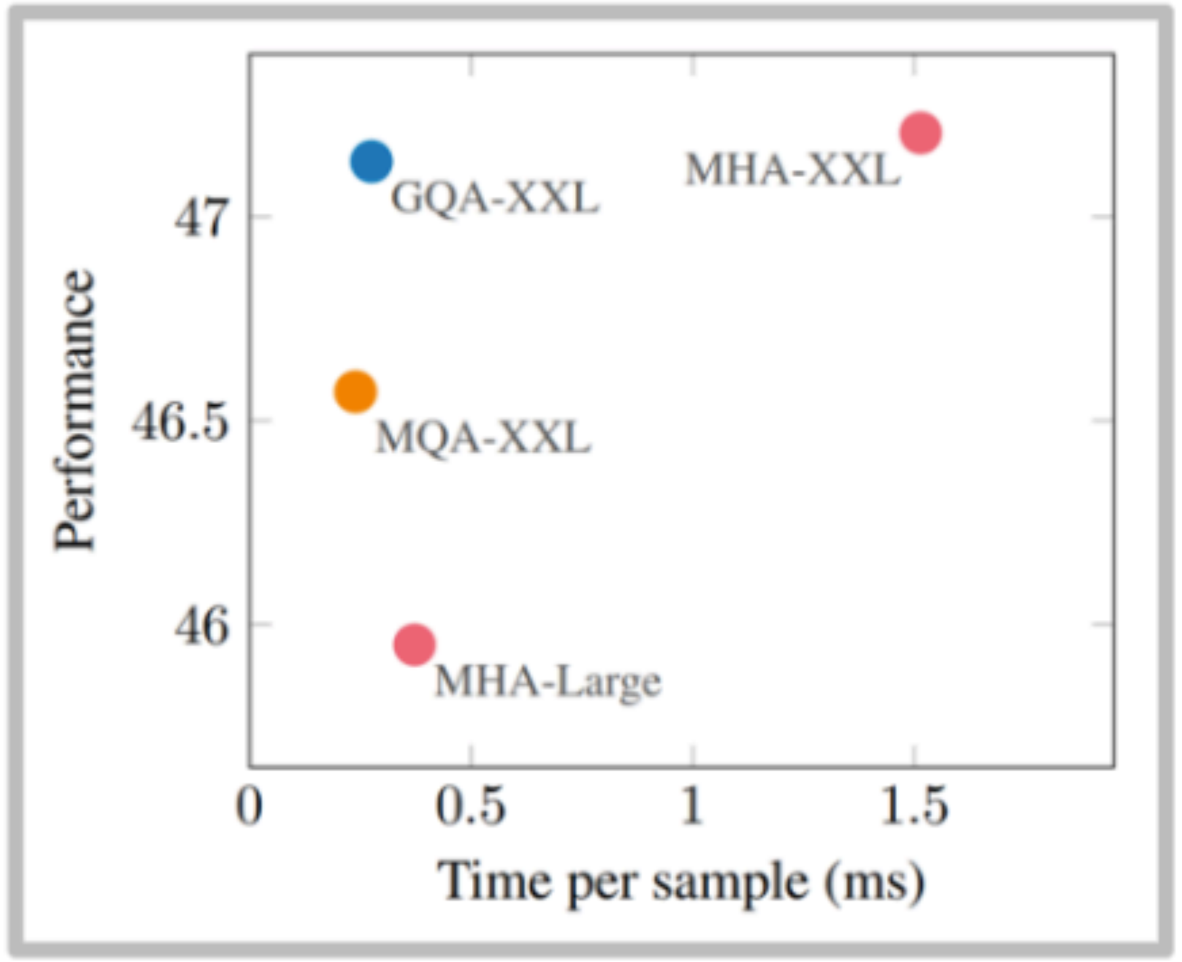
Does MQA/GQA hurt?

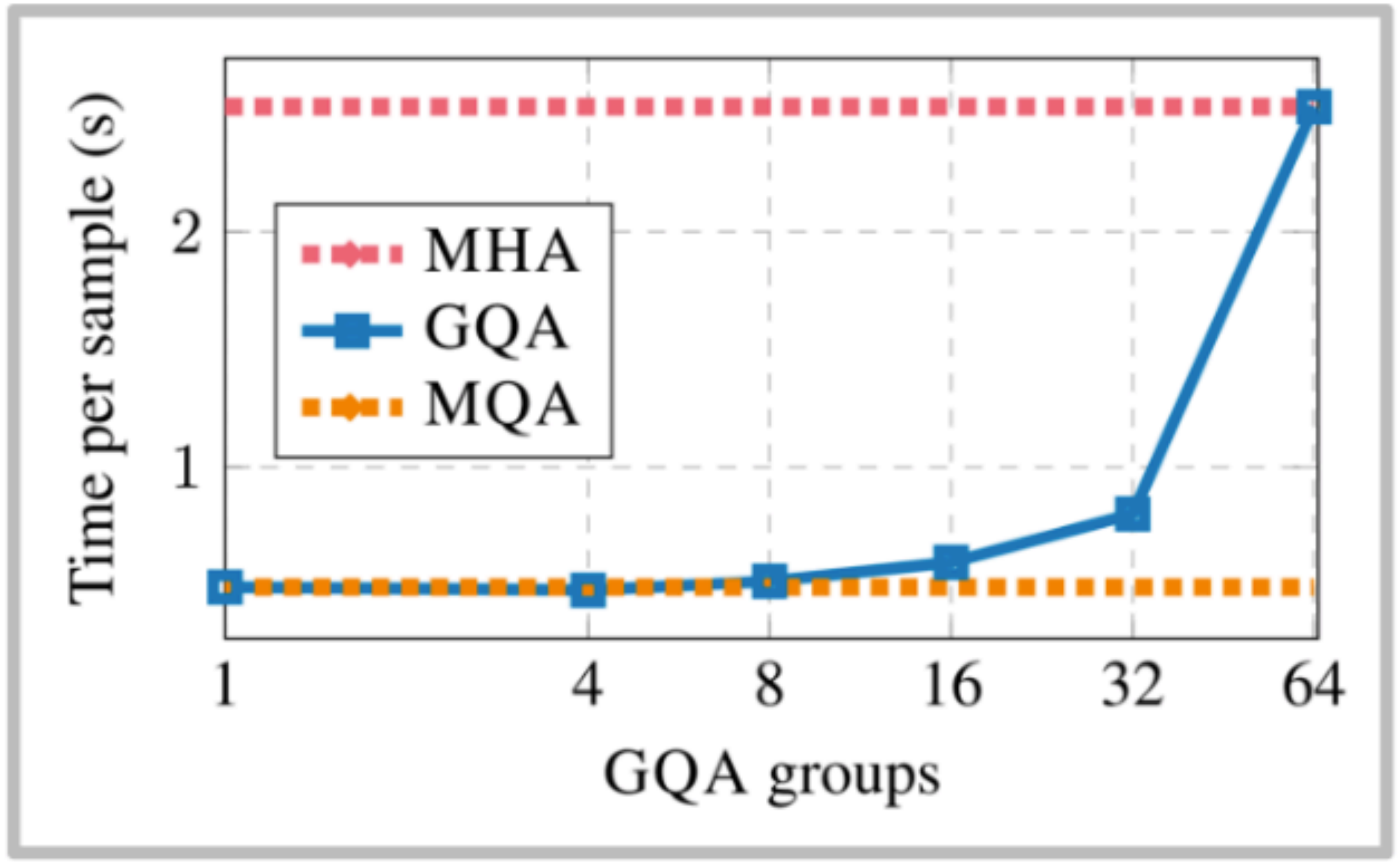
Small PPL hit w/ MQA [Shazeer 2019]

Table 3: Billion-Word LM Benchmark Results.

Attention	h	d_k, d_v	d_{ff}	dev-PPL
multi-head	8	128	8192	**29.9**
multi-query	8	128	9088	30.2
multi-head	1	128	9984	31.2
multi-head	2	64	9984	31.1
multi-head	4	32	9984	31.0
multi-head	8	16	9984	30.9

Low/no hit w/ GQA [Ainslie 2023]





Q&A